

Instruction Interpretation & Sequencing (Self-Learning)

Q. Explain the concepts of Instruction Interpretation and Sequencing. How is the control flow of instructions maintained during execution

[DEF] Definition to Remember

Program execution relies on two mechanisms: Instruction Interpretation (the Control Unit decoding the instruction to understand *what* to do) and Instruction Sequencing (the process of using the Program Counter to determine *which* instruction to execute next).

1. Instruction Interpretation

The process where the Control Unit determines the meaning of a fetched machine instruction.

2. Instruction Sequencing

The process of determining the memory address of the *next* instruction to fetch.

3. Modifying Control Flow

While the default flow is sequential, programs require loops and decisions. Control Flow Instructions alter the sequencing process by changing the PC:

4. Fetch-Decode-Execute Cycle

The synthesis of both concepts:

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Instruction Interpretation: The Control Unit decodes the Opcode to understand the operation.
2. It generates specific hardware control signals based on the interpretation.
3. Instruction Sequencing: Determines the address of the next instruction to fetch.
4. Sequencing relies heavily on the Program Counter (PC).
5. Default execution is Sequential (PC is automatically incremented after fetch).
6. Control flow is altered by Branch Instructions (Conditional or Unconditional) which manually overwrite the PC.
7. For subroutines, the old PC is pushed to a stack to maintain the eventual sequential return path.

[RECALL] Quick Recall

Interpretation (Decode Opcode, generate signals) -> Sequencing (Find next address via PC) -> Default: $PC = PC + 1$ -> Control Flow: Branches change the PC to jump/loop